Surgery 174 (2023) 1008–1020

Contents lists available at ScienceDirect

Surgery

journal homepage: www.elsevier.com/locate/surg

Presented at Academic Surgical Congress 2023

The gap between hospital-based violence intervention services and client needs: A systematic review

Angie Jang, BS[a], Arielle Thomas, MD, MPH[b], John Slocum, MPH[a], Kaithlyn Tesorero, BS[c], Giovanna Danna, BS[d], Anjay Saklecha, BS[a], Eileen Wafford, MSt, MLIS, AHIP[e], Sheila Regan, BA[f], Anne M. Stey, MD, MSc, FACS[a,*]

[a] Feinberg School of Medicine, Northwestern University, Chicago, IL
[b] Department of Surgery, Medical College of Wisconsin, Milwaukee, WI
[c] University of Illinois College of Medicine at Chicago, Chicago IL
[d] Chicago Medical School, Rosalind Franklin University, North Chicago, IL
[e] Galter Health Sciences Librarian and Learning Center, Feinberg School of Medicine, Northwestern University
[f] Acclivus, Inc, Chicago, IL

ARTICLE INFO

Article history:
Accepted 8 July 2023
Available online 14 August 2023

ABSTRACT

Background: Survivors of intentional interpersonal violence face social challenges related to social determinants of health that led to their initial injury. Hospital-based violence intervention programs reduce reinjury. It is unclear how well they meet clients' reported needs. This systematic review aimed to quantify how well hospital-based violence intervention program services addressed clients' reported needs.

Methods: Medline, The Cochrane Library, CINAHL Plus with Full Text, and PsycInfo were queried for studies addressing hospital-based violence intervention programs services and intentional injury survivors' needs in the United States. Case reports, reviews, editorials, theses, and studies focusing on pediatric patients, victims of intimate partner violence, or sexual assault were excluded. Data extracted included program structure, hospital-based violence intervention program services, and client needs assessments before and after receiving hospital-based violence intervention program services.

Results: Of the 3,339 citations identified, 13 articles were selected for inclusion. Hospital-based violence intervention programs clients' most reported needs included mental health (10 studies), employment (7), and education (5) before receiving hospital-based violence intervention programs services. Only 4 studies conducted quantitative client needs assessments before and after receiving hospital-based violence intervention program services. All 4 studies were able to meet at least 50% of each of the clients' reported needs. The success rate depended on the need and program location: success in meeting mental health needs ranged from 65% to 90% of clients. Conversely, time-intensive long-term needs were least met, including employment 60% to 86% of clients, education 47% to 73%, and housing 50% to 71%.

Conclusion: Few hospital-based violence intervention programs studies considered clients' reported needs. Employment, education, and housing must be a stronger focus of hospital-based violence intervention programs.

© 2023 Elsevier Inc. All rights reserved.

Introduction

Disadvantaged minorities, especially young Black males from a low socioeconomic background, have been disproportionately affected by intentional interpersonal violence.[1] This reflects ingrained racial disparities in social determinants of health, including safe housing, education, and employment.[2–4] The violence reinjury rate for survivors varies from 16% to 44%.[4–9] The major contributing factor has been that violence survivors have been discharged to identical social conditions that led to the initial injury. They face the same social determinants of health needs before the injury and new post-injury needs, such as accessing follow-up medical care and legal concerns.[10,11] Survivors also have dealt with new physical disability, financial difficulties, and mental

This work was presented as an oral presentation at the Academic Surgical Congress, February 9, 2023, in Houston, TX.

* Reprint requests: Anne M. Stey, MD, MSc, FACS, Department of Surgery, Northwestern University, 676 N. St. Clair, Suite 650, Chicago, IL 60611.
E-mail address: as013j@gmail.com (A.M. Stey);
Twitter: @ac_thomas7, @AnneMStey

https://doi.org/10.1016/j.surg.2023.07.011
0039-6060/© 2023 Elsevier Inc. All rights reserved.

health consequences that have increased distrust and isolation from their communities.[12]

Hospital-based violence intervention programs (HVIPs) have addressed clients' social determinants of health needs to reduce re-injury.[13–17] These programs have intervened during the "teachable moment" of hospitalization from an intentional interpersonal violent injury where survivors have engaged with the healthcare system. Hospital-based violence intervention programs have prevented violent re-injury by engaging with survivors through peer-based case management using hospital-based resources and referring to community-based services.[18,19] There has been great heterogeneity between HVIPs.[19] Numerous studies have assessed HVIPs and determined effectiveness based on reduction in violence re-injury rates.[19] However, no systematic literature review has assessed effectiveness from clients' perspectives. Hospital-based violence intervention program effectiveness should be based on how well they meet clients' reported needs.

The aim of this systematic literature review was to answer the following PICOS (Population, Intervention, Comparison, and Outcome) question: among survivors of intentional interpersonal violence (participants) who received HVIP services postdischarge (intervention), what were their self-reported social determinants of health needs (outcome), and how well were these needs met as captured in observational studies and qualitative studies (study design). The secondary aim was to assess the HVIP structures and services that enabled programs to better meet clients' reported needs.

Methods

This review was performed and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.[20] The protocol for this systematic review was prospectively registered with PROSPERO (CRD42021267762).

Eligibility criteria

The following inclusion criteria were established to answer the PICOS question: (1) P, randomized control trial, observational cohort, case-control, or qualitative studies; (2) I, survivors of intentional interpersonal violence that received post-discharge services from HVIP; (3) C, analysis of clients' reported needs and HVIP services; (4) O, study setting in the United States and article originally published in English; (5) C, publication date after the year 2000. Report characteristics used as eligibility criteria were English-language studies and, when available, a United States geographic filter. These eligibility criteria were applied given the unique organized trauma system in the United States and its downstream effects on social determinants of health. Furthermore, given the limited relevant articles found before 2000, the publication date was restricted to after 2000. There was no predefined duration of study follow-up because follow-up was client-tailored. Case reports, reviews without original data, editorials, and theses were excluded. Studies focused only on pediatric patients, domestic violence or intimate partner violence survivors, or sexual assault survivors were excluded. This limit was determined because this was often the target population for most HVIPs.[1,21]

Information sources

MEDLINE (Ovid), The Cochrane Library (Wiley), CINAHL Plus with Full Text (Ebsco), and PsycInfo (Ebsco) were leveraged as information sources. These wide-ranging information sources were used because HVIP studies have been documented commonly in nursing, social services, and psychology literature, given the multi-disciplinary nature of the underlying HVIP work and the social determinants of health the programs seek to address. References of included studies were reviewed to identify additional relevant citations through purposive sampling.

Search strategy and data collection

A comprehensive search strategy was developed with a research librarian (Q.E.W.). The search strategy combined Medical Subject Headings and keywords for victims, families, HVIP, and intentional interpersonal injury (Supplementary Tables S1–S3). The original search was conducted on September 29, 2021, by the research librarian (Q.E.W.). One author (A.J.) manually rescreened for newer studies on January 27, 2023.

After completing the literature search, 4 independent reviewers (A.J., A.T., J.S., K.T.) screened the titles and abstracts based on the above-predetermined protocol inclusion and exclusion criteria. Duplicates and those not meeting inclusion criteria were excluded. The remaining articles were reviewed independently in full length by 5 reviewers (A.J., A.T., J.S., K.T., G.D.). At least 2 reviewers assessed every article independently. Any discrepancies between reviewers during the entire screening were resolved through consensus or the involvement of a third reviewer. Finally, the references of the included articles were checked for additional articles of interest through purposive sampling.

Data items

Five reviewers (A.J., A.T., J.S., K.T., G.D.) independently extracted data. At least 2 reviewers assessed data for extraction in every article. Any discrepancies between reviewers during the entire screening were resolved through consensus or the involvement of a third reviewer. Data item variables for extraction included details about the HVIP (name, location, client inclusion criteria, types of services available, timeline of follow-up), study design (study sample size, study participant demographics, measured outcomes, and important results), client (quantitative or qualitative) reported needs before and after HVIP services, and study limitations. Data items were extracted into a Google Excel Sheet. A full list of data items was reviewed by the full study team (Supplementary Digital Content S2).

Summary measures

The primary summary measures were clients' reported needs before and after HVIP services. Secondary summary measures were unique program interventions or the HVIP structure that enabled programs to better meet clients' reported needs. The studies ' investigators obtained or confirmed further data or confirmation of data beyond what was printed in the published text.

Risk of bias and study quality assessment

The validated Methodological Index for Non-Randomized Studies criteria was used to assess the quality of the included observational articles (Supplementary Digital Content S3).[22] The risk of bias was evaluated for each observational study based on 12 items. Each item was scored from 0 (not reported), 1 (reported but inadequate), or 2 (reported and adequate). The ideal total score for noncomparative studies was 16 and 24 for comparative studies.

Data synthesis

Results were synthesized qualitatively, given the limited number of high-quality clinical articles and the great methodological

heterogeneity between HVIPs and study measures. Meta-analysis was not appropriate or attempted.

Results

Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines, the searches identified 3,339 overall references. After removing duplicate studies (n = 724), 2,596 unique references were screened. In addition, 2,557 articles were excluded after title/abstract screening, and 40 studies were assessed for eligibility by full-length review. A total of 13 articles (12 through systematic search and 1 through manual search) discussing a broad range of clients' reported needs were selected for inclusion in the review (Figure 1).[9,10,23–32]

Bias assessment

Studies had a wide range of study sizes, from 11 to 839 participants, with a follow-up period ranging from 6 to 12 months (Table I). All studies were qualitative (53.8%) or observational (46.2%). There were no randomized controlled trials that fit the inclusion criteria. The Methodological Index for Non-Randomized Studies criteria of risk of bias within the 6 quantitative non-comparative studies was a median of 11.5 and ranged from 9 to 12. Six programs were represented in this review from unique programs (Table II).

Results of individual studies demonstrated that the most frequently identified client-reported needs beyond medical care were mental health, employment, education/vocational training, legal services, financial/victim-of-crime assistance, housing, and immigration services. Only 4 of the 13 studies evaluated had client-reported needs assessments before and after receiving HVIP services.[9,10,23,30] The most successfully met client-reported needs included mental health, legal services, and financial/victim-of-crime assistance. The least met needs included employment, education, housing, and help with obtaining driver's licenses. All 4 studies were able to meet at least 50% of each of the identified needs.

Mental health

A total of 9 studies identified mental health as the most common need among HVIP clients before HVIP services. HVIP clients described post-traumatic stress disorder symptoms (eg, hypervigilance, social isolation, intense fear, emotional detachment) and signs of anxiety and depression.[10,26,27,31,32] Richardson et al found that this also contributed to the distrust and hesitancy toward using hospital resources.[26] Thus, meeting mental health-related needs should be a priority: clients were 6 times more likely to successfully

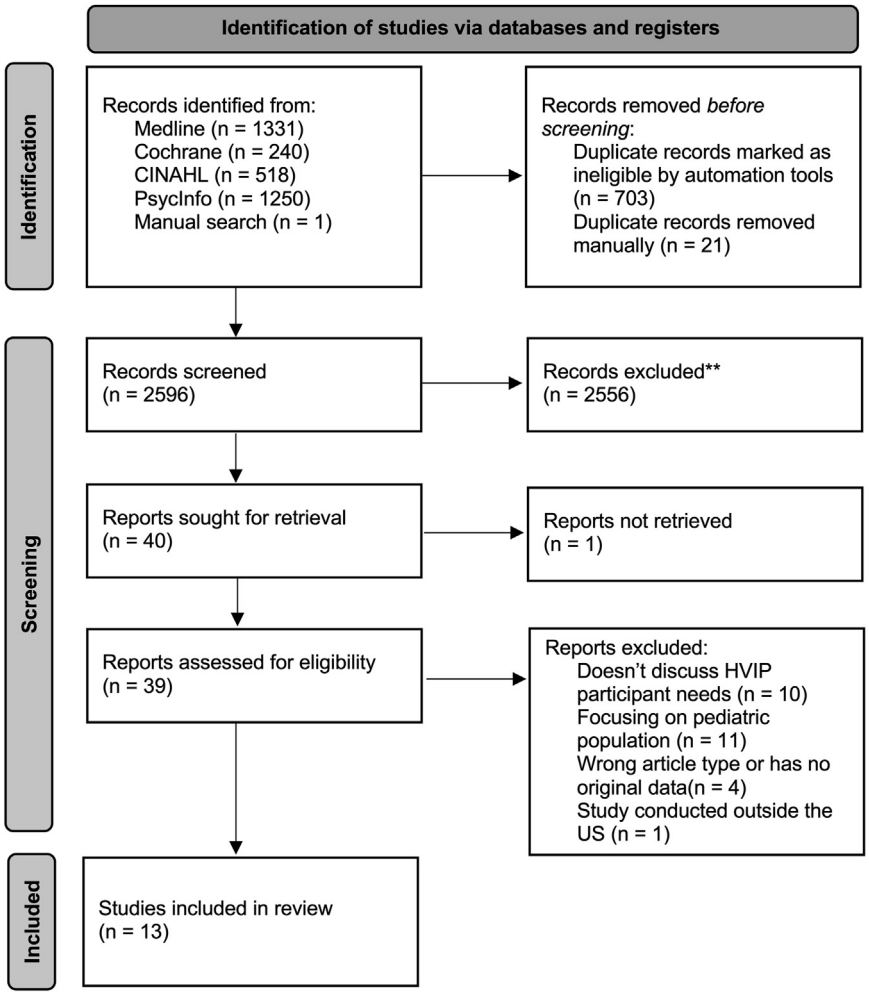

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow diagram for study identification, screening, and inclusion. *HVIP*, hospital-based violence intervention program.

Table I
Summary of included articles

Authors	Name of HVIP	Study design	Participant demographics (%)		Major findings	Needs-related outcomes	
Dicker et al[23] (2009)	Wraparound Project	Retrospective cohort study Analyzing client data during first year of program (2005–2006) 45 enrolled clients between 12–30 Level of Evidence: III	African American Latino Pacific Islander White Male Female	59% 29% 4% 7% 86% 14%	Of 435 people meeting eligibility, 54% had identified needs Violence prevention should incorporate community input when making strategies of risk reduction Six to 7 major needs were successfully addressed in at least half of the population with greater needs of basic services and education 6-mo attrition rate for program was 4% 90% of clients were unemployed at time of injury	Identified need Court advocacy Education Vocational training Mental health/drug treatment Employment Housing Driver license	Needs met, % 88% 68% 67% 65% 61% 50% 14%
Smith et al[9] (2013)	Wraparound Project	Retrospective cohort study Analyzing client data between Jan 2005 to Dec 2011 254 enrolled clients between 10–30 y old Level of Evidence: III	African American Hispanic White Asian Native American Other/multiple Unknown Male Female	59% 26% 5% 2% 0.04% 4% 3% 87% 13%	Of 254 clients, 234 identified needs during initial assessment Meeting mental health and employment needs had the largest impact on community violence 4.5% reinjury rate for clients receiving services	Identified need Other Court advocacy Mental health Vocational/professional training Family counseling Education Housing Employment Driver's license Substance abuse	Needs met, % 52/58 (89%) 53/62 (85%) 86/111 (80%) 36/50 (72%) 20/30 (66%) 54/83 (65%) 54/86 (62%) 54/91 (59%) 19/41 (46%) 2/5 (40%)
Juillard et al[10] (2016)	Wraparound Project	Retrospective cohort study Analyzing prospectively collected data between Feb 2005–Sep 2015 Clients enrolled before Sep 30, 2014 459 enrolled clients between 10–35 y old Level of Evidence: III	Black/African American Latino White Other/multiple Male Female	47% 44% 5% 5% 86% 14%	11.3% overall reinjury rate over 10 y Increased risk of reinjury by having needs in housing, education, court advocacy, and driver's license Meeting needs for VOC and education needs was significantly protective of future re-injury Successful in meeting needs especially if served by community-based organizations with dedicated services	Identified need VOC Visa Other Tattoo removal Court advocacy Mental health Family counseling Education Housing Employment Driver's license	Needs met, % 181/188 (96%) 35/37 (95%) 55/61 (90%) 8/9 (89%) 76/87 (87%) 172/202 (85%) 30/37 (81%) 81/110 (73%) 86/119 (72%) 88/143 (62%) 22/42 (52%)

(*continued on next page*)

Table I (*continued*)

Authors	Name of HVIP	Study design	Participant demographics (%)		Major findings	Needs-related outcomes
Patel et al[24] (2020)	Wraparound Project	Qualitative study 15 semi-structured interviews with clients, friends and families, case managers, and other stakeholders	Clients Case managers Family members/ friends of clients Program director Licensed clinical social worker	47% 28% 13% 7% 7%	Human-centered design study that revealed case managers desire clear boundaries with patients, but patients desire closer relationships Development of an Information and Communication Technology application aims to meet the needs of clients while supporting case managers to efficiently offer their services	Case manager/client relationship: A successful relationship is both professional and personal. Clients need this around the clock, but case managers can only support this while on the clock Communications need to feel personal, but they do not always need to be personalized Healing is a journey of skill development and lifestyle changes that must be acknowledged, monitored, and rewarded Social networks need to provide peer support for healing rather than peer pressure to propagate violence
Decker et al[25] (2020)	WAP	Qualitative study 24 participants who were WAP clients, completed 6 mo of intensive case management and were 18+ Both clients with neutral to positive relationships and clients with uniquely strong relationships with case managers	African American Latino White Other/multiple Male Female	58% 44% 5% 5% 88% 13%	Strong client-case manager relationships were heavily linked to clients' success in program HVIPs should prioritize community involvement and cultural concordance when hiring case managers Clients appreciated case managers that are dependable, accessible, and reliable during the long-term	Characteristics of case managers appreciated by clients: Understand and relate to client sociocultural context similar experiences, share a racial/gender identity, or open to learning Engaging with friends/family to build trust Exhibit true compassion and care by "going 'the extra mile' to provide services" (eg, driving them to medical appointments, sitting with them at court, etc) Act as role models that can be a formal and informal mentor ("a man I want to look up to") Serve as portals of opportunity and engender mutual respect/ pride
Richardsonet al[26] (2020)	CG-VIP	Qualitative study Semi-structured interviews with 11 young Black male survivors of violent firearm injury participating in HVIP 18–30 y old Participants had history of criminal justice involvement Focus group of 6 participants	Black White Hispanic Other/multiple Male Female	75% 6% 12% 6% 87% 13%	Participant experiences of posttraumatic stress symptoms included hypervigilance, avoidance, sleep disturbance, irritability, isolation and distrust, irritability Important to consider interpersonal violence (firearm) and structural violence (incarceration) in understanding survivors' mental health HVIP was unable to evaluate its effectiveness due to high rates of missed attendance Telehealth model may address concerns for in-hospital services	Hypervigilance may reduce ability to receive services, suggesting a potential preference for receiving off-site services Self-imposed forms of social isolation and general may hinder relationship with violence intervention specialist Group therapy facilitated by a psychotherapist and violence intervention specialist was helpful for survivors Caregivers and the violence intervention specialist important in recovery
Richardson et al[27] (2021)	CG-VIP	Qualitative study 11 qualitative individual interviews with clients between 18–35 y old who engaged with CG-VIP and were hospitalized for at least 1 gunshot wound at least once Focus group with 4 of the 11 to explore role of CG-VIP	African American Male	100% 100%	HVIP participation is related to quality of service and financial burden of access DNEMT services perceived as safer than traditional public transportation due to greater sense of personal safety and no engagement with violence Using Uber Health over traditional rideshare transportation streamlined communication and transparency	Transportation costs were major barrier to accessing psychosocial services; even though program was seen positively, many clients would not have participated because of financial barrier of transportation Free DNEMT service was a primary factor for engagement for many clients DNEMT services increased adherence to appointments with probation officers, increasing program credibility

(*continued on next page*)

Table I (*continued*)

Authors	Name of HVIP	Study design	Participant demographics (%)		Major findings	Needs-related outcomes	
Wical et al[28] (2020)	CAP-VIP	Qualitative study Interviews with 10 young Black male survivors of gun violence with history of criminal justice involvement that interacted with VIS Focus groups (2) each with 6 different participants Interview with credible VIS	African American Male	100% 100%	Among 116 participants of CAP-VIP, 1% trauma reinjury rate Is having shared experiences with participants is crucial VIS can use both nonclinical and clinical perspective to serve as a male role model Beyond clinical terms, it is necessary to consider needs and viewpoints of survivors in defining credibility	Use of a relational rather than a transactional model of engagement helps establish trust with clients Importance of shared experience: ability to share personal experiences were essential in helping the clients feel understood When clients' credibility and trust for the VIS was developed, clients' resistance to therapy decreased VIS could act as a male role model (which is often missing for clients) due to shared lived experiences	
Foran et al[29] (2020)	Los Angeles County Hospital's Tattoo Removal Program in partnership with community-based organization	Retrospective observational study Clients recruited to by case manager and active clients 81 clients (18–64 y old) completed anonymous survey	N/A		100% of respondents considered tattoo removal to be a part of an overall transition to a healthier and safe lifestyle Motivations for engagement: improve employment opportunities, improve mental health, or for personal safety Tattoos may serve as identifying factor for current or former gang members, which may increase chances for confrontations with rival gangs or police Free tattoo-removal services tattoo removal may be a useful adjunct for victims of interpersonal violence seeking a healthier and safer life	Client satisfaction in meeting motivations for tattoo removal: Improve employment opportunities Improve safety/avoid violence Leave gang affiliation Removal of tattoo related to abusive intimate partner	 39/48 (81%) 19/24 (80%) 16/17 (94%) 12/14 (86%)
Pino et al[30] (2021)	Boston VIAP	Retrospective cohort study Analyzed data of HVIP clients with complete client engagement and services data between 2013–2018 Of 2,243 identified survivors, 839 engaged with services Level of Evidence: III	Black Hispanic White Other Unknown Male Female	73% 12% 11% 3% 1% 85% 15%	Significant predictors of VIAP engagement: younger, black race, permanent housing, existing mental health diagnosis, penetrating injury due to firearm, and severe injuries Needs that were more likely to be successfully achieved within one year were basic needs, injury recovery, and medical/mental health needs Education, employment, housing, legal, and family support needs have lower probability of goal achievement compared with basic needs VIAP successful in meeting >80% of client-identified needs	Identified need Medical and mental health Basic needs Injury recovery Family Legal Housing Employment Education	Needs met, % 1,852/2,058 (90%) 593/681 (87%) 1184/1390 (85%) 380/486 (78%) 750/987 (76%) 534/753 (71%) 211/351 (60%) 97/208 (47%)

(*continued on next page*)

Table I (*continued*)

Authors	Name of HVIP	Study design	Participant demographics (%)		Major findings	Needs-related outcomes
James et al[31] (2014)	Boston VIAP	Qualitative study HVIP clients enrolled between July 1, 2010 –June 30, 2011 who were English speaking adults at least 18 y old presenting to the ED with penetrating trauma were eligible Randomly selected clients to recruit Semistructured interviews with 20 clients	Black Hispanic/other Male Female	70% 30% 70% 30%	Client experiences with the intervention advocate evolved from initial distrust to connection through communication building and mutual trust One client felt relationship did not last because the violence intervention advocate did not push enough; nonetheless, many described long-term relationships with the advocate Advocates assisted with accessing counseling and support, help with education, employment, and other life skills, as well as building meaningful and lasting relationships with their clients. The peer support from the HVIP and talking to a caring adult made clients perceive the program as positive and effective	Major challenges for survivors were symptoms of PTSD, isolation, lack of trust, fear, and safety Multiple clients described longer term relationships with violence intervention advocate that went beyond physical recovery, a gap often missed by other service providers Counseling and support were the most commonly cited services Advocates helped survivors by providing emotional support and addressing challenges including those related to education, employment, and life skills
Hink et al[32] (2022)	HIFI	Qualitative study 16 HIFI participants between 20–54 y old at least a year after enrollment were interviewed Interviewer was blinded to enrollment arm Interviews conducted during 7-mo period in 2019	Black White Asian Hispanic Multiple Male Female	25% 56% 6% 6% 6% 63% 38%	Clients express significant mental health sequelae following injury including symptoms of PTSD, difficulty forming relationships, etc Formal and informal social networks (family and friends, employers) were imperative for support Difficulty navigating mental health provider access, compensation paperwork and hospital bills, as well as were major barriers to recovery Mental health assessment, connections to financial support services, mitigating legal challenges, and support in finding a sense of mission are potential opportunities for HVIPs Providing education and referrals for safe firearm ownership may support survivors	Many participants discussed the importance of addressing mental health needs; those with ongoing psychological support stressed how important it was for recovery Difficulty finding mental health providers was common and some gave up on actively pursuing services Financial assistance, reduced payment plans, and state crime victims' compensation were valuable to recovery Many expressed disappointment with the criminal justice system due to poor communication, few updates, and sense of apathy toward individual case Most participants were employed or in school prior to injury, but expressed employment gave a sense of purpose and value Return to independence was a major need for survivors

(*continued on next page*)

Table I (*continued*)

Authors	Name of HVIP	Study design	Participant demographics (%)		Major findings	Needs-related outcomes
Gorman et al[33] (2022)	HVIP at University Hospital	Arm 1: Retrospective cohort analysis of 259 HVIP participants (July 1, 2017–Oct 1, 2020) to analyze patient goals Arm 2: Adult English speaking, nonincarcerated Black/Hispanic male survivors not in program interviewed as control Arm 3: Semi-structured qualitative interviews with HVIP patients	Black Other Unknown White American Indian Male Female	82% 11% 5% 2% 0.3% 86% 14%	49% of HVIP participants reached stated goals within 6 mo Engagement with HVIP associated with decreased PTSD and higher positive affect HVIP clients more likely to achieve early positive health outcomes 30% of patients had early disengagement (<2 post discharge HVIP encounters) Participants expressed that communication with case manager, who served as formal and informal mentor, was frequent and necessary	49% participants achieved stated goal within 6 mo of hospital discharge 51% achieved one goal; 36% achieved 2 stated goals; 9% achieved 3 goals Most commonly achieved goals were medical, VOC, emergency food services Most commonly provided services were emergency funds, access to information, and counselling/mentoring Long-term goals (eg, relocation, insurance, education) were more difficult to achieve

CG-VIP, County General Violence Intervention Program; *DNEMT*, digital nonemergency medical transportation services such as Uber Health; *ED*, emergency department; *HIFI*, Helping Individuals with Firearm Injuries; *HVIP*, hospital-based violence intervention program; *PTSD*, post-traumatic stress disorder; *VIAP*, Violence Intervention Advocacy Program; *VOC*, victims-of-crime assistance; *VIS*, Violence Intervention Specialist; *WAP*, Wraparound Project.

complete their program when mental health needs were met.[9] The success rate for meeting mental health needs among the 4 studies that conducted needs assessments before and after HVIP services ranged from 65% to 90%.[9,10,23,30] Cultural stigma represented a major barrier to clients seeking mental health care, but several studies found that cultivating a stronger relationship between the client and case manager reduced it.[9,10,23] Furthermore, several studies identified long wait times and transportation costs as additional barriers to using mental health services.[10,27,32]

Employment

A total of 6 studies identified employment as a major client-reported need. According to Foran et al, a majority of their clients who opted for tattoo removal cited improving employment opportunities as their main motive.[29,30] In addition to the financial stability provided by employment, 2 studies noted how these resources also gave them a sense of purpose.[25,32] Even among clients that were already employed, getting appropriate work accommodations to keep their job after the injury from their employer played a significant role during recovery.[32] Although employment was a common and important need, programs found it more difficult to meet. All 4 quantitative studies highlighted employment as one of the relatively least met needs.[9,10,23]Employment was still important to meet as Smith et al found that if employment needs were met for a client, they were 4 times more likely to be "successful" in their program (judged as having at least 50% of their initial needs met).[9]

Education/vocational training

The success rate in meeting educational needs ranged from 46.6% to 72.7%.[9,10,23,32] Although Juillard et al reported that education and vocational/professional training were one of the better-met needs from the Wraparound Project (73% of identified needs met), education needs were more difficult to meet for other programs.[24,29,33] Hospital-based violence intervention programs heavily relied on community-based organizations and partnerships with local schools to help meet education needs through tutoring support, General Educational Diploma training, and vocational training, so the availability of such resources shaped how well programs could meet these needs. Nonetheless, meeting education needs were significantly protective against future re-injury.[10]

Legal services

Pino et al found that legal needs were reduced by 15% after HVIP services.[30] Success in meeting legal needs ranged from 76% to 88% of clients. One study found that having needs in education and legal needs predicted an increased risk of re-injury.[10] James et al also found that distrust toward the legal system prevented clients from building new, healthy relationships.[31] Overall, studies found that frustrations with the legal process drove this need due to poor communication of the status of their injury investigations, little help in finding or convicting assailants, and a perceived sense of apathy.[32] Dicker et al found that case managers could play an important supervisory role in working with clients and important members of the legal system to facilitate a greater dialogue.[23]

Financial/victim-of-crime assistance

Studies found that their clients highly valued reduced payment plans provided by the hospital in addition to state crime victims' compensation to help alleviate the financial barrier during recovery.[10,34] Only Juilliard et al directly reported their success rate in

Table II
Summary of represented HVIPs

Program	Location	Client inclusion criteria	Engagement and follow-up process	Select services provided
Wraparound Project[9,10,23,24]	San Francisco General Hospital	High-risk survivors of violent trauma (defined as gunshot, stabbing, blunt assault) 10–30 y old Live or injured in San Francisco Excluded victims of child abuse, domestic violence, or self-inflicted	Trained case managers identify high-risk potential clients within 24 h of presenting to ED Case manager engages at bedside and conducts initial risk assessment If eligible offered enrollment and case manager conducts intake and needs assessment Follow-up generally 6 mo to 1 y Client graduates when major needs are met	Crisis response services Mental health services for victim and family Cognitive Behavioral Therapy Higher education Art workshops Substance abuse services Youth programs Housing/relocation Court advocacy School support Employment Vocational training After school programs with Department of Parks Referrals to community-based organizations Assistance with Driver's License Tattoo removal program
HVIP at University Hospital[33]	UH in Newark, NJ	Survivors of interpersonal violence (gunshot, stabbing, assault) >18 y old	Culturally competent trauma case managers approach survivors at bedside and offer services Needs assessment performed at consent and design plan of action Case managers schedules meetings at participant's home, court, or community locations and help refer to resources Case manager meets with patient daily during inpatient hospitalization, 1–2× weekly for few weeks after discharge, and every 3 m for next year Participants graduate when needs are met	Select services include: Case worker makes referrals to services for: Mental heath Career Education Case worker provides advocacy and accompanies during: Medical care Medical forensic exams Law enforcement interviews Case worker aids with: Shelter/housing Emergency funds ≤$200 Victims of crime paperwork
Capital Region Violence Intervention Program[26–28]	UM Prince George's Hospital Center	Youth and adults with violent injuries	The Credible Messenger typically recruits at hospital bedside or over the phone Clients receive information from program director and then if choose to engage Social worker/clinical counselor, case manager, and Credible Messenger work together Follow up generally 6–12 mo	Services include: Counseling (individual and support groups) Mental health counseling for caregivers Intensive case management services Job training, life skills, and job placement Education/vocational training Housing and legal assistance Health insurance coverage Free transportation to services Referrals to community-based organizations
Violence Intervention Advocacy Program[30,31]	BMC	Survivors of violent penetrating trauma (gun shot or stab wound) treated at BMC ED >15 y old	VIA recruited from Boston communities and trained VIA contact clients at bedside or through phone calls Advocates conduct initial needs assessment and develop individual plans Advocates then maintain case management relationships after discharge with clients choosing to engage Home visiting nurse travels once per week There is no limit to amount of time client can receive services, but on average, participation is around 1 y	Provide immediate crisis intervention, safety planning, and resources for survivors and families Promote trauma recovery and provide support for needs including: Safety Mental health Life skills training (eg, financial literacy) Housing Education Legal support Primary care Employment Medical insurance Family support services to family members affected Recreation

(*continued on next page*)

Table II (*continued*)

Program	Location	Client inclusion criteria	Engagement and follow-up process	Select services provided
Helping Individuals with Firearm Injuries[32]	Harborview Medical Center (Seattle, WA)	Survivors of firearm assault and unintentional injury English speaking >18 y old Residing in a 5 county area around Seattle Not under arrest Able to provide consent within 4 wk of hospital discharge Discharge home from initial admission	RCT: 6-mo intervention program with primary aims of reducing incarceration and violence reinjury Support Specialist conducts motivational interviewing to increase engagement Upon enrollment, clients receive 6 mo of regular contacts by the Support Specialist to identify relevant recovery resources needed	Worked with multidisciplinary team consisting of local community organizations, law enforcement agencies, and public service agencies to address needs including: Mental health Substance abuse Legal support Education Access to food Employment Domestic violence
TRP[29]	Los Angeles County Hospital	Anyone willing to perform requisite community service Survivors of violence receive referrals from hospital	Clients recruited by dedicated case manager or other active clients Free tattoo removal sessions provided in exchange for 5 h of community service (eg, active participation in community intervention programs) In partnership with Soledad Enrichment Achievement and Gang Reduction and Youth Development	Tattoo removal services (part of larger violence prevention and community outreach initiative)

BMC, Boston Medical Center; *ED*, emergency department; *HVIP*, Hospital-Based Violence Intervention Programs; *RCT*, randomized controlled trial; *TRP*, Tattoo Removal Program; *UH*, University Hospital; *VIA*, violence intervention advocates.

meeting financial/victim-of-crime assistance, which was 96.3%.[10] They also found that providing financial/victim-of-crimes assistance was protective of future re-injury. As previously mentioned, costs were a major barrier that prevented clients from engaging with services. Richardson et al found that providing free transportation services with Uber Health reduced this barrier to HVIP participation, allowing more clients to remain engaged with the HVIP.[27]

Housing

The studies reporting quantitative needs assessments reported that 50% to 72% of identified housing needs were met.[9,10,23,30] Housing was consistently among the least met needs because finding affordable and safe housing was labor intensive given that options were limited. Although Boston's HVIP attempted to pilot a housing option for its homeless clients, the results were mixed.[30]

Tattoo removal services

Two studies discussed the needs addressed by tattoo removal services.[10,29] Juillard et al reported that 37 clients used this service, and the need was satisfied 95% of the time.[10] Foran et al found that most clients opted for services to improve employment opportunities and personal safety, goals that were met for >80% of clients.[29]

Obtaining drivers' licenses

Only the Wraparound Project reported its success in meeting needs related to obtaining a driver's license.[9,10,23] This need had one of the lowest success rates, which was attributed to secondary barriers, including illiteracy.

Case management enabled HVIP to better serve clients needs

The HVIP structure that enabled programs to better meet clients' reported needs was the use of case managers with lived experience with interpersonal violence. Two of the 13 assessed studies specifically evaluated the estimated dose-response of contact with the case managers. Nine of the 13 studies described the case manager and client-assisted dissemination of the program services. Four studies involved clients explicitly describing the case managers' influence on participation in HVIP. Case managers were important for 3 reasons. First, case managers were regarded as credible messengers who legitimized the program. Most case managers came from the community and had a personal understanding of client needs. This understanding could be derived from openness and willingness to learn or, more commonly, from personal experiences with violence. This connection was consistently noted to be an essential part of the development of a relationship that leads to participation and continuation in the program. When case managers demonstrated empathy, compassion, care, and respect to their clients, they were able to foster trust with the client. Multiple studies mentioned how showing commitment by providing employment assistance and directly bringing clients to resources was especially important in building trust between clients and case managers.[9,25,32] They served as role models and examples of the potential for successful change. Second, case managers served as a bridge between the social care system and the client. They served as patient navigators within the complex medical system. Smith et al showed that in the first 3 months, clients who received 3 to 6 hours per week of exposure to case managers were almost 5 times more likely to be successful in meeting their needs than clients who received 0 to 1 hour.[9] Clients who received >6 hours of exposure were 5.6 times more likely to be successful. Last, the case managers

were extremely important to the evaluation of the HVIP. Recruitment for evaluating client-reported needs was done through case managers as an initial point of contact in 3 of the included studies.

Finally, a unique communication platform was piloted by one program to better serve clients' reported needs. Patel et al identified the communication needs of clients aged 16 to 30 years old with case managers at the San Francisco Wraparound Project to assess a mobile Information Communication Technology tool currently in use.[24] This analysis underscored a tension between client needs and the case manager's capacity to inform potential strategies to improve client-case manager communication. Interviews revealed the potential for poorly designed tools to negatively affect the safety of clients and case managers alike.

Discussion

Intentional interpersonal violence survivors have been shown to be almost twice as likely to get violently injured again within 2 years compared with their control counterparts with nonviolent injuries.[34] Hospital-based violence intervention programs can address survivors' social determinants of health. Gorman et al found that compared with survivors who did not engage in services, HVIP clients were much more likely to return to work or school and comply with medical follow-up.[33] In this systematic review, HVIP services mostly addressed certain clients' reported needs. The most frequently identified client needs included mental health, employment, education, legal services, and financial/victim-of-crime assistance. The best-met needs were mental health care, legal services, and financial/victim-of-crime assistance. The least met needs were employment, education, housing, and attaining driver's licenses. The Victims of Crime Advocacy and Recovery Program in Cleveland, Ohio found similar needs, citing education as the most used services among 1,019 program clients.[35] Ultimately, a comprehensive, collaborative approach is necessary to addressing educational needs and empowering survivors. Mueller et al found that establishing this community-hospital-university network between community partners, hospital staff, and academic researchers increased program success.[36]

Overall, meeting client needs related to social determinants of health can have a great impact. Survivors and their families often face long-term mental health consequences, including post-traumatic stress disorder symptoms and increased isolation from communities that, if gone unaddressed, were suggested to increase the risk for violent victimization.[37,38] Furthermore, financial instability has been identified as an underlying driver for community violence, which may be reduced by increasing quality education and employment.[39] Educational programs were important in increasing access to meaningful work, especially for youth that did not complete high school.[29] Currently, most HVIPs use the reduction of violent re-injury and incarceration as the main objectives to gauge program success. However, these measures may be limited due to small sample sizes, loss to follow-up, and unaccounted readmissions outside of the trauma centers associated with the HVIPs.[11,33] Given the complex influences related to determinants of health affecting social interventions like HVIPs, especially when survivor safety can be affected, observational qualitative studies may be more appropriate and practical than randomized control trials in providing insights into previously unrecognized clients' needs.[40] Meeting underlying social determinants of health needs would also impact future reinjury, so a greater program focus on meeting these needs would be more valuable.

Among a variety of client services, meeting long-term needs like education, employment, and housing was more difficult due to limitations in local resources and funding and the barrier of high attrition in many programs. Floyd et al found that barriers to retention in their intervention included challenges in communication with the client due to client housing instability, overworked staff, and misaligned work schedules between clients and case managers.[41] However, distrust between clients and the healthcare system may be another major contributing factor. Survivors of violence often describe their time in the hospital as impersonal and dehumanizing, which Rich et al attribute to the hyper-routinization of trauma care.[42,43] Other studies have also found that distrust of health care providers among gun violence survivors was a barrier to using mental health services.[44] Thus, building the VI-client relationship is important for increasing trust. Clients stated that when case managers had an openness to learn and were more culturally informed, the relationship between clients and HVIPs was improved.[45] Voith et al conducted focus groups and semi-structured interviews of HVIP staff and found that building relational health by involving families and hospital staff also showed case managers' investment, which helped to increase trust.[46] Given that low participation has been a critical barrier to HVIP effectiveness, understanding what barriers limit client engagement is necessary for adapting HVIP programming. In this review, the cost of transportation and hypervigilance after injury were identified as 2 potential secondary barriers. Focused research on client-reported needs-based outcomes and fulfillment may support such endeavors.[47]

Study limitations

This systematic review had several limitations. First, very few studies conducted client needs assessments and the studies that did represented only a handful of HVIP programs. This may have led to bias through the representation of the most successful programs. Second, HVIP resources were constrained by the availability of community-based organizations and resources. These local variations contribute to the differences in HVIP program structure and unique adaptations required for each program to address local client needs. As such, the generalizability of our study findings may be limited to individual local contexts. Third, the quality and type of the evidence were generally limited. Most study designs had selection bias likely driving outcomes, where the most motivated clients participated. Furthermore, although there have been a few randomized controlled trials pertaining to HVIP efficacy in the past, they have mainly been limited by small sample sizes or significant loss to follow-up, and none focused on needs-based outcomes.[14,16,21] Fourth, the outcome of interest, clients reported needs, was self-reported. This may have led to response or recall bias. A final fifth limitation was the small sample sizes of most studies. A recent systematic review considering HVIP effectiveness on re-injury had similar limitations.[20]

In conclusion, overall, our review found that HVIP services addressed clients reported needs for mental health, employment, education/vocational training, legal services, and financial/victim-of-crime assistance >50% of clients (Table III). However, employment, education, and housing remain important opportunities, which must be addressed collaboratively to build trust and long-term continuity of client support. Hospital-based violence intervention programs must continue to foster a collaborative approach between community and hospital partners to address the client's needs and focus more on client-reported needs and patient-centered outcomes rather than re-injury alone.

Funding/Support

This work did not receive any direct financial support. A.M.S. was funded by the American Association for the Surgery of Trauma, the American College of Surgeons, and the National Institutes of

Table III
Important takeaways

Select review findings	Considerations
Mental health was the most frequently identified need, but long wait times, transportation costs, and distrust for health care providers may be barriers preventing clients from using mental health resources	Potential strategies for addressing these barriers may include: • using in-program mental health providers, off-site services, or telehealth services • strategies to addressing financial barriers including provision of free digital nonemergency medical transportation services to mental health providers • strengthening the relationship with the VIS.
Clients often felt great distrust toward the judicial system, especially with the lack of communication about individual investigations and general apathy that made them feel unheard. Thus, support with legal services was a common need	• HVIPs can help address these needs by establishing ties with the local judicial system and engaging with judges, public defenders, and probation officers to increase transparency. • Case managers may play an important supervisory role to connect clients with the legal system and build rapport with both parties.
Long-term needs, such as employment, education, and housing were more difficult to meet. Employment and education were among the most frequently identified needs, but also 2 of the least met needs for several programs	Attrition is a major barrier for HVIPs that make it harder to meet long-term needs. • A close relationship with the VIS is important to building trust with the client. VIS with shared experiences and empathy reflected in sincere commitment (eg, driving clients to services rather than just referring) were especially successful at serving as a role model and patient navigator. • Community-based organizations and partnerships with local institutions were especially important in meeting long-term needs, so establishing partners early on was important. • Engaging family members and community members helped to increase familiarity with the program. Secondary barriers, such as transportation barriers and distrust towards the health care system, unique to the local client population may prevent the usage of services. • More needs-based outcomes research is necessary to gauge what needs are not as well addressed among survivors. • Qualitative interviews may be valuable in providing insight on secondary barriers. • Using patient-tailored interventions to address these barriers, such as free ride share program and a mobile app, may help boost communication.

HVIP, hospital-based violence intervention program; *VIS*, Violence Intervention Specialist.

Health/National Heart Lung and Blood Institute (K23 HL157832–01).

Conflict of interest/Disclosure

The authors have no conflicts of interests or disclosures to report.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [https://doi.org/10.1016/j.surg.2023.07.011].

References

1. Centers for Disease Control and Prevention, National Center for Injury Prevention and Control. Web-Based Injury Statistics Query and Reporting System (WISQARS). Available from: https://www.cdc.gov/injury/wisqars/index.html. Accessed July 1, 2010.
2. Haider AH, Young JH, Kisat M, et al. Association between intentional injury and long-term survival after trauma. *Ann Surg*. 2014;259:985–992.
3. Peek-Asa C, Butcher B, Cavanaugh JE. Cost of hospitalization for firearm injuries by firearm type, intent, and payer in the United States. *Inj Epidemiol*. 2017;4:20.
4. Kao AM, Schlosser KA, Arnold MR, et al. Trauma recidivism and mortality following violent injuries in young adults. *J Surg Res*. 2019;237:140–147.
5. Sims DW, Bivins BA, Obeid FN, Horst HM, Sorensen VJ, Fath JJ. Urban trauma: a chronic recurrent disease. *J Trauma*. 1989;29:940–947.
6. Morrissey TB, Byrd CR, Deitch EA. The incidence of recurrent penetrating trauma in an urban trauma center. *J Trauma*. 1991;31:1536–1538.
7. Goins WA, Thompson J, Simpkins C. Recurrent intentional injury. *J Natl Med Assoc*. 1992;84:431–435.
8. Crandall CS, Jost PF, Broidy LM, Daday G, Sklar DP. Previous emergency department use among homicide victims and offenders: a case-control study [published correction appears in *Ann Emerg Med*. 2005;45:220] *Ann Emerg Med*. 2004;44:646–655.
9. Smith R, Dobbins S, Evans A, Balhotra K, Dicker RA. Hospital-based violence intervention: risk reduction resources that are essential for success. *J Trauma Acute Care Surg*. 2013;74:976–982.
10. Juillard C, Cooperman L, Allen I, et al. A decade of hospital-based violence intervention: benefits and shortcomings. *J Trauma Acute Care Surg*. 2016;81:1156–1161.
11. Bell TM, Gilyan D, Moore BA, et al. Long-term evaluation of a hospital-based violence intervention program using a regional health information exchange. *J Trauma Acute Care Surg*. 2018;84:175–182.
12. Wiggall S, Boccellari A. The UC San Francisco Trauma Recovery Center Manual: a model for removing barriers to care and transforming services for survivors of violent crime. Available from: http://traumarecoverycenter.org/wp-content/uploads/2020/04/TRC-Manual.pdf. Accessed August 18, 2022.
13. Cunningham R, Knox L, Fein J, et al. Before and after the trauma bay: the prevention of violent injury among youth. *Ann Emerg Med*. 2009;53:490–500.
14. Zun LS, Downey L, Rosen J. The effectiveness of an ED-based violence prevention program. *Am J Emerg Med*. 2006;24:8–13.
15. Becker MG, Hall JS, Ursic CM, Jain S, Calhoun D. Caught in the crossfire: the effects of a peer-based intervention program for violently injured youth. *J Adolesc Health*. 2004;34:177–183.
16. Cooper C, Eslinger DM, Stolley PD. Hospital-based violence intervention programs work. *J Trauma*. 2006;61:534–540.
17. Purtle J, Dicker R, Cooper C, et al. Hospital-based violence intervention programs save lives and money. *J Trauma Acute Care Surg*. 2013;75:331–333.
18. Johnson SB, Bradshaw CP, Wright JL, Haynie DL, Simons-Morton BG, Cheng TL. Characterizing the teachable moment: is an emergency department visit a teachable moment for intervention among assault-injured youth and their parents? *Pediatr Emerg Care*. 2007;23:553–559.
19. Affinati S, Patton D, Hansen L, et al. Hospital-based violence intervention programs targeting adult populations: an Eastern Association for the Surgery of Trauma evidence-based review. *Trauma Surg Acute Care Open*. 2016;1:e000024.
20. Moher D, Liberati A, Tetzlaff J, Altman DG, PRISMA Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *BMJ*. 2009;339:b2535.
21. Aboutanos MB, Jordan A, Cohen R, et al. Brief violence interventions with community case management services are effective for high-risk trauma patients. *J Trauma*. 2011;71:228–237.
22. Slim K, Nini E, Forestier D, Kwiatkowski F, Panis Y, Chipponi J. Methodological index for non-randomized studies (minors): development and validation of a new instrument. *ANZ J Surg*. 2003;73:712–716.
23. Dicker RA, Jaeger S, Knudson MM, et al. Where do we go from here? Interim analysis to forge ahead in violence prevention. *J Trauma*. 2009;67:1169–1175.
24. Patel D, Sarlati S, Martin-Tuite P, et al. Designing an information and communications technology tool with and for victims of violence and their case managers in San Francisco: human-centered design study. *JMIR Mhealth Uhealth*. 2020;8, e15866.
25. Decker HC, Hubner G, Nwabuo A, et al. "You don't want anyone who hasn't been through anything telling you what to do, because how do they know?": qualitative analysis of case managers in a hospital-based violence intervention program. *PLoS One*. 2020;15, e0234608.
26. Richardson Jr JB, Wical W, Kottage N, Bullock C. Shook ones: understanding the intersection of nonfatal violent firearm injury, incarceration, and traumatic

stress among young Black men. *Am J Mens Health*. 2020;14, 1557988320982181.
27. Richardson Jr JB, Wical W, Kottage N, Galloway N, Bullock C. Staying out of the way: perceptions of digital non-emergency medical transportation services, barriers, and access to care among young black male survivors of firearm violence. *J Prim Prev*. 2021;42:43–58.
28. Wical W, Richardson J, Bullock C. A credible messenger: the role of the violence intervention specialist in the lives of young Black male survivors of violence. *Violence and Gender*. 2020;7:66–69.
29. Foran CP, Clark DH, Alling T, et al. Laser tattoo removal: a new tool for hospital-based violence prevention? *Am Surg*. 2020;86:1337–1344.
30. Pino EC, Fontin F, James TL, Dugan E. Boston violence intervention advocacy program: challenges and opportunities for client engagement and goal achievement. *Acad Emerg Med*. 2021;28:281–291.
31. James TL, Bibi S, Langlois BK, Dugan E, Mitchell PM. Boston Violence Intervention Advocacy Program: a qualitative study of client experiences and perceived effect. *Acad Emerg Med*. 2014;21:742–751.
32. Hink AB, Atkins DL, Rowhani-Rahbar A. Not all survivors are the same: qualitative assessment of prior violence, risks, recovery and perceptions of firearms and violence among victims of firearm injury. *J Interpers Violence*. 2022;37: NP14368–NP14396.
33. Gorman E, Coles Z, Baker N, et al. Beyond recidivism: hospital-based violence intervention and early health and social outcomes. *J Am Coll Surg*. 2022;235: 927–939.
34. Cunningham RM, Carter PM, Ranney M, et al. Violent reinjury and mortality among youth seeking emergency department care for assault-related injury: a 2-year prospective cohort study. *JAMA Pediatr*. 2015;169:63–70.
35. Simske NM, Rivera T, Ren BO, et al. Implementation of programming for survivors of violence-related trauma at a level 1 trauma center. *Trauma Surg Acute Care Open*. 2021;6, e000739.
36. Mueller KL, Moran V, Anwuri V, Foraker RE, Mancini MA. An exploration of factors impacting implementation of a multisystem hospital-based violence intervention program. *Health Soc Care Community*. 2022;30: e6577–e6585.
37. Vella MA, Warshauer A, Tortorello G, et al. Long-term functional, psychological, emotional, and social outcomes in survivors of firearm injuries. *JAMA Surg*. 2020;155:51–59.
38. Orcutt HK, Erickson DJ, Wolfe J. A prospective analysis of trauma exposure: the mediating role of PTSD symptomatology. *J Trauma Stress*. 2002;15: 259–266.
39. Halimeh BN, Hughes D, Evans B, et al. Empowering the affected: informing community-based solutions through interviews with survivors of interpersonal firearm violence: perspectives of survivors of firearm injuries. *J Trauma Acute Care Surg*. 2021;90:980–986.
40. Ranjan S, Neudecker CH, Strange CC, et al. Hospital-based violence intervention programs (HVIPs): making a case for qualitative evaluation designs. *Crime & Delinquency*. 2023;69:487–509.
41. Floyd AS, Lyons VH, Whiteside LK, Haggerty KP, Rivara FP, Rowhani-Rahbar A. Barriers to recruitment, retention and intervention delivery in a randomized trial among patients with firearm injuries. *Inj Epidemiol*. 2021;8: 37.
42. Patton D, Sodhi A, Affinati S, Lee J, Crandall M. Post-discharge needs of victims of gun violence in Chicago: a qualitative study. *J Interpers Violence*. 2019;34: 135–155.
43. Rich JA. *Wrong Place, Wrong Time: Trauma and Violence in the Lives of Young Black Men*. Baltimore, MD: Johns Hopkins University Press; 2009.
44. O'Neill KM, Vega C, Saint-Hilaire S, et al. Survivors of gun violence and the experience of recovery. *J Trauma Acute Care Surg*. 2020;89:29–35.
45. Hosking JE, Ameratunga SN, Bramley DM, Crengle SM. Reducing ethnic disparities in the quality of trauma care: an important research gap. *Ann Surg*. 2011;253:233–237.
46. Voith LA, Lee H, Salas Atwell M, et al. A phenomenological study identifying facilitators and barriers to Black and Latinx youth's engagement in hospital-based violence intervention programs. *Health Soc Care Community*. 2022;30: e4873–e4884.
47. Monopoli WJ, Myers RK, Paskewich BS, Bevans KB, Fein JA. Generating a core set of outcomes for hospital-based violence intervention programs. *J Interpers Violence*. 2021;36:4771–4786.